



# Scope of Work - Enterprise Virtualization Platform Replacement (VMware Alternative)

**RFP Alternative Solutions and Migration Support**

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**Orange Group**

13 May 2025

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# 1 Introduction

Orange Group (hereinafter referred to as "Client") is issuing this Request for Proposal (RFP) to solicit proposals from qualified vendors (hereinafter referred to as "Vendor") to select and implement a comprehensive, scalable, resilient, and cost-effective enterprise virtualization platform, including integrated management, Software-Defined Networking (SDN), and Disaster Recovery (DR) orchestration capabilities, to **fully replace** the Client's current extensive and complex existing VMware environment.

This RFP seeks a comprehensive software platform solution intended to run on Client's existing or future commodity hardware. The solution must encompass the core virtualization platform, integrated management, and critically, robust Software-Defined Networking (SDN) and Disaster Recovery (DR) orchestration capabilities equivalent to the Client's current deployment. A primary driver for this initiative is addressing VMware licensing cost and model concerns. Therefore, proposals must clearly articulate a path to significant Total Cost of Ownership (TCO) reduction.

Given the scale and complexity detailed herein, a critical component of the proposal must be a detailed, proven methodology, tooling description, and service offering for migrating diverse and demanding workloads from the current VMware environment (spanning multiple versions and advanced features) to the proposed platform with minimal disruption.

## 2 Project Goals, Drivers

### 2.1 Project Drivers

- **Primary Driver:** To select and implement a comprehensive, scalable, resilient, and cost-effective enterprise virtualization platform, including integrated management, Software-Defined Networking (SDN), Infrastructure as Code (IaC), API availability, automation and Disaster Recovery (DR) orchestration capabilities, to **fully replace** the Client's current VMware environment.
- Address significant challenges and achieve cost efficiencies related to **current and anticipated VMware licensing models and costs**.
- Desire to evaluate modern platform alternatives offering flexibility, potential for open standards alignment, and predictable long-term economics.
- Need for a platform capable of supporting a large, complex, multi-generational VMware environment consolidation.
- Requirement for robust, integrated SDN and DR orchestration capabilities within the core platform offering.

## 2.2 Project Goals

- **Top Objective: Significantly reduce the Total Cost of Ownership (TCO)** for the enterprise virtualization platform over a 3 and 5-year period compared to the projected costs of remaining on the incumbent VMware stack. Proposals must quantify the expected savings.
- Successfully migrate most existing workloads (from the environment detailed in [Section 4](#)) to the new platform, ensuring stability and performance for demanding applications **(including Telco workloads, large databases, VDI, SAP, CRM, billing systems)** while adhering to defined migration windows.
- Implement a highly scalable and resilient platform capable of managing infrastructure at the Client's scale with simplified, efficient operations.
- Achieve functional parity for critical capabilities currently provided by:
  - VMware NSX (Data Center / NSX-T)
  - VMware Site Recovery Manager (SRM) Enterprise
- Implement equivalent multi-tenant/cloud capabilities if required based on current vCD, vRA or in house developed tools usage.
- Maintain or improve infrastructure agility and automation capabilities.

## 3 In/Out Scope Deliverables & Services

### 3.1 Proposed Platform Deliverables

#### 3.1.1 Core Platform Software

- Provision of the proposed virtualization/infrastructure platform software licenses/subscriptions designed to run on Client's existing x86 hardware infrastructure.
- Software must be suitable for deployment on Client's existing/future commodity x86 hardware as specified in Section 4, unless vendor is proposing an option.

#### 3.1.2 Integrated Management Software

- Provision of licenses/subscriptions for the centralized management platform capable of efficiently managing the environment at the specified scale and meeting requirements in [Section 5](#).

#### 3.1.3 Integrated SDN Capabilities

- Provision of licenses/subscriptions for a native SDN solution providing functional parity with the Client's current usage of NSX Data Center / NSX-T (as detailed in [Section 4](#) and required in [Section 5](#)). This includes micro-segmentation, routing, load balancing, etc.

### **3.1.4 Integrated DR Orchestration Capabilities**

- Provision of licenses/subscriptions for a native DR orchestration solution providing functional parity with the Client's current usage of VMware SRM Enterprise (as detailed in [Section 4](#) and required in [Section 5](#)) and capability to manage other solutions outside of vendor's stack. This includes recovery plan management, non-disruptive testing, failover/failback automation.

### **3.1.5 Integrated Operations Management Capabilities**

- Provision of integrated monitoring, log collection, capacity planning, basic automation, and/or multi-tenant cloud management capabilities or possibility to use other solutions outside of vendor's stack.
- Provision of robust platform monitoring and reporting, with clear integration points for Client's existing third-party monitoring tools.
- Provision of centralized management software/tools capable of operating at the specified scale.

### **3.1.6 Integrated Cloud Management / Automation**

- Provision of robust APIs with CLI, integration points (Ansible, Terraform) enabling Client to build necessary automation workflows and portal for simple service consumption or management.

### **3.1.7 Platform Documentation**

- Provision of comprehensive technical documentation (installation, administration, API guides, best practices) in English.

## **3.2 Migration Services [CRITICAL]**

### **3.2.1 Assessment and Planning Services**

- Conduct a thorough assessment of the current VMware environment (using Vendor tools and Client data).
- Deliver detailed dependency mapping for applications and infrastructure components.
- Provide compatibility analysis and identify any remediation required before migration.
- Develop a comprehensive, phased Migration Plan, including wave scheduling, prerequisite identification, resource requirements (Vendor and Client), communication plan, rollback strategy, and risk assessment, tailored to the Client's scale and complexity (including multiple vSphere versions, NSX configurations, SRM plans, potentially vCD constructs) to the proposed platform.

### **3.2.2 Migration Tooling and Execution Services**

- Provide and manage the necessary V2V migration tools capable of handling the required migration types (cold, warm/live) and scale, as detailed in.

- Ensure migration facilities and compatibility with environments already used in Orange as for example sylvia framework.
- Execute the migration of VMs according to the agreed Migration Plan and schedule.
- Manage bulk migration processes efficiently and reliably.
- Provide regular progress reporting and manage exceptions/failures during execution.

### **3.2.3 Network & Security Configuration Migration Assistance**

- Provide tools and/or expert services to assist in translating and implementing equivalent network configurations (NSX segments, routing, LB pools) and security policies (NSX DFW rules) onto the new platform's SDN solution while maintaining the assigned IP addresses.

### **3.2.4 DR Plan Migration Assistance**

- Provide tools and/or expert services to assist in recreating and configuring equivalent DR recovery plans within the new platform's DR orchestration solution.

### **3.2.5 Validation and Cutover Services**

- Define and execute post-migration validation procedures (technical checks, potentially assisting with Client application testing).
- Manage the cutover process for each migration wave according to the plan.
- Provide dedicated hypercare/support during critical cutover windows.
- Provide post-migration platform support and maintenance services.

## **3.3 Implementation & Training Services**

### **3.3.1 Platform Implementation Services**

- Install, configure, and validate the proposed virtualization platform, management tools, SDN, and DR components according to agreed design and best practices. This should include the capability of building POC / demo to evaluate operational efforts.

### **3.3.2 Training Services**

- Deliver technical training to Client's administrators and operations staff covering platform administration, management, troubleshooting, SDN management, and DR operations, as detailed in [Section 8](#).

### **3.3.3 Support Services**

- Provide ongoing technical support and maintenance services for the newly implemented platform according to the SLAs and procedures defined by the Client with minimum of 24x7x365 support for severity 1 incidents.
- Professional services for platform implementation and configuration.

### 3.4 Explicitly Out of Scope:

- Procurement, installation, and maintenance of physical server hardware, storage hardware, or network hardware (unless vendor is proposing a specific, optional integrated hardware/appliance solution).
- Application development, modification, refactoring, or application-level performance tuning. Migration focus is on the VM/infrastructure layer.
- Management, patching, or support of Guest Operating Systems (Windows, Linux) or any applications running *inside* the virtual machines.
- Physical data center facilities management (space, power, cooling, cabling).
- Management or decommissioning of legacy VMware licenses or support contracts.
- Data backup *execution* (Vendor provides API/integration, Client's backup team/tool performs backups), unless Managed Backup is proposed as an optional service.

### 3.5 Optional Services and capabilities

- Data backup execution (Vendor provides API/integration, Client's backup team/tool performs backups), unless Managed Backup is proposed as an optional service.
- Managed Services: Ongoing operational management of the new platform (beyond standard break/fix support).
- Ongoing Platform Optimization Services: Regular performance tuning, capacity optimization reviews post-migration.
- Custom Integration Development: For specific needs not covered by standard APIs/integrations.
- Solution enabling capacity to manage and move VM on public or private cloud environment

### 3.6 Key assumptions and dependencies

- Client will provide accurate information about the current environment.
- Client will provide necessary access (physical and logical) to current and future environments for assessment, migration, and implementation.
- Client technical staff will be available for knowledge transfer, collaboration during planning, validation testing, and required training.
- Client will ensure necessary network connectivity (bandwidth, latency) between source VMware environment, target platform, and migration tools/appliances.
- Client is responsible for Guest OS and Application-level compatibility testing and validation post-migration.



- Client is responsible for procuring the necessary underlying hardware meeting the Vendor's HCL requirements (for software-only deployments).

### 3.7 Proof of Concept (PoC)

The client anticipates requiring shortlisted vendors to conduct a PoC. Proposals should detail the Vendor's standard PoC process, duration, resource requirements, and key validation objectives they would demonstrate, particularly concerning migration, SDN, DR, and performance for demanding workloads.

### 3.8 Use Case Guidance

This section includes use cases intended to illustrate some examples of the Client's requirements.

#### 3.8.1 Use Case 1: Migrating Micro-segmented Application

- **Scenario:** "A critical 3-tier application (Web, App, DB) currently relies on NSX DFW rules for micro-segmentation between tiers. The migration must maintain or enhance this security posture on the new platform's SDN."
- **Required Vendor Focus:** Describe the specific tools/process for discovering existing NSX rules, translating them into the target SDN policy format, implementing the new policies, and validating firewall enforcement before and after the VMs are migrated. How is rule consistency ensured during the migration window?

#### 3.8.2 Use Case 2: Migrating Telco NFV Component with DR

- **Scenario:** "A latency-sensitive Telco Network Function VM requires specific resource guarantees (CPU pinning, NUMA alignment) and is protected by an SRM recovery plan with a sub-1-hour RPO/RTO."
- **Required Vendor Focus:** Detail how the migration tool preserves/re-applies performance-critical VM settings. Describe how the corresponding DR plan will be recreated on the new platform's DR solution, including non-disruptive testing validation before migrating the production VM, ensuring RPO/RTO objectives can still be met.

#### 3.8.3 Use Case 3: Large Database Cluster Migration

- **Scenario:** "A multi-TB Microsoft SQL Server Failover Cluster Instance (FCI) or Oracle RAC cluster runs on VMware, requiring minimal downtime during migration."
- **Required Vendor Focus:** Describe the specific live/warm migration technique suitable for large, high-change-rate database VMs. Detail coordination needed with DBAs, pre/post-migration checks specific to clustered databases, and performance validation strategy to ensure equivalent or better performance on the new platform. How are shared disks (RDMs or VMDKs) handled?

## 4 Proposal Structure Requirements

- **The Vendor shall submit three (3) comprehensive proposals**, each including both **commercial** and **technical** details, corresponding to different levels of required CPU core capacity. The proposals shall be structured as follows:
  - **Offer 1:** Based on a total of **30,000 cores**
  - **Offer 2:** Based on a total of **50,000 cores**
  - **Offer 3:** Based on a total of **100,000 cores**
- Additionally, each proposal must explicitly explain two (2) items that are:
  - The proposed solution regarding our expectations
  - The Migration possibilities and services to master the risk and reduce the cost
- The **100,000-core figure represents the global consolidated volume**, aggregated at Group level through localized purchasing activities.
- For further information regarding the existing VMware environment and the technical requirements for the proposed solution, Vendors shall refer to:
  - **Section 5 – VMware Environment Overview**
  - **Section 6 – Technical Requirements for Proposed Solution**
- **Orange reserves the right to award contracts to one or more Vendors** for the provision of services under this RFP, at its sole discretion and based on strategic, operational, or financial considerations.

## 5 Current VMware Environment Overview (to be replaced)

This section details the current VMware environment that is the subject of this replacement and migration RFP. The information provided represents the approximate scale and complexity across the Client's participating affiliates. Vendors must propose solutions capable of replacing this environment and migrating workloads effectively.

### 5.1 VMware Products in Scope for Replacement

The table below summarizes the VMware products that are included in the scope of this RFP.

| Product Type   | Versions                                                              |
|----------------|-----------------------------------------------------------------------|
| vCenter Server | Essentials 4, 5, 6, 7, 8<br>Foundation 4, 6<br>Standard 4, 5, 6, 7, 8 |

|                         |                                                                                                                                                       |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| vCenter Site            | Enterprise 5, 6, 8                                                                                                                                    |
| Recovery Manager        |                                                                                                                                                       |
| vCloud Director         |                                                                                                                                                       |
| vCloud Suite            | Advanced 5, 6<br>Enterprise 5, 6                                                                                                                      |
| NSX Data Center         | Advanced per Processor<br>Enterprise Plus per Processor                                                                                               |
| NSX-T                   | Enterprise Plus                                                                                                                                       |
| vRealize Automation     | Advanced 7<br>Standard 7                                                                                                                              |
| vRealize Operations     | Advanced 8<br>Enterprise 6, 7, 8<br>Standard 8                                                                                                        |
| vRealize Suite          | Advanced 7, 2017, 2018, 2019<br>Enterprise 7, 2017, 2019<br>Standard 7, 2018, 2019                                                                    |
| vSphere                 | Advanced 4<br>Enterprise 4, 5, 6, 7<br>Enterprise Plus 5, 6, 7, 8<br>Essentials 4, 5, 6, 7, 8<br>Essentials Plus 4, 5, 6, 7<br>Standard 4, 5, 6, 7, 8 |
| vSAN                    | Advanced 6, 6.1, 6.2, 7, 8<br>Enterprise 6, 7<br>Standard 7                                                                                           |
| Workstation             | 15 Pro                                                                                                                                                |
| VMware vDefend Firewall |                                                                                                                                                       |
| Avi Load Balancer       |                                                                                                                                                       |

## 5.2 Environment Scale

| Inventory Global Data Information                                               |                                            | Average number of VMs |
|---------------------------------------------------------------------------------|--------------------------------------------|-----------------------|
| Approximate Total Cores                                                         | 100,000 cores                              | -                     |
| Average number of cores per server for affiliates for VMware vSphere Foundation | 16 or 18 cores / CPU with 2 CPU per server | 23                    |

|                                                                                     |                                         |    |
|-------------------------------------------------------------------------------------|-----------------------------------------|----|
| Average number of cores per server<br>for affiliates for VMware Cloud<br>Foundation | 26 cores / CPU with 2<br>CPU per server | 33 |
|-------------------------------------------------------------------------------------|-----------------------------------------|----|

## 5.3 Physical Locations & Time Zones

Primary operational regions requiring support coverage span European (EU) and Middle East & Africa (MEA) time zones.

## 5.4 Underlying Hardware

The preferred hardware platform to utilize is listed below:

- Primary Server Vendors:
  - DELL
    - 13G / 14G / 15G / 16G rack servers R6x0
    - Blade servers MX7x0c
    - PowerEdge M630 / M640 / M640 blade / MX740c / R640 / R650 / R740 / R750 / R760 / 740XD / R660 vSAN Ready
  - HPE
    - ProLiant DL380 / DL385 / DL360 / BL460c / DL580 / BL685c / Gen 8 / 9 / 10 / 10Plus
    - Synergy 480 / G10 / G11
  - Ericsson
  - IBM
    - X3650 M4
    - X3850 X5
  - Huawei
- Primary Storage Vendors:
  - DELL
    - Powerstore
    - Metronade
    - PowerMax
    - VMAX
    - EMC Unity
  - NetApp

- ASA90
  - FAS2750
- HPE
  - Alletra
  - Primera
  - MSA 2060
- Pure Storage
  - FlashArray
- Ericsson
  - Nexenta
- Huawei
- Current Support Arrangement: Currently under VMware Production Support

## 5.5 Storage

Primary storage types: FC SAN, iSCSI SAN, NFS NAS, vSAN.

## 5.6 Networking Configuration

- Virtual Switching: Primarily utilizes vSphere Standard Switch (VSS), vSphere Distributed Switches (VDS), NSX.
- Network Segmentation: Extensive use of VLANs. Significant reliance on NSX for logical networking and micro-segmentation.
- IP Addressing: Primarily IPv4 with expected capabilities for IPv6

## 5.7 Backup, Disaster Recovery & Operations

- Backup: Primarily uses Veeam Backup & Replication and Commvault, utilizing e.g. image-level, agent-based backups.
- Disaster Recovery: Relies heavily on VMware SRM Enterprise with [Specify primary replication] for orchestrated recovery. Stated RPO/RTO targets for critical applications are typically [Specify targets, e.g., RPO < 1 hour, RTO < 4 hours].
- Monitoring: Utilizes native platform tools (vCenter Alarms, potentially Aria Ops) supplemented by third-party tools including: BMC Event Manager, Patrol, Nagios, Zabbix, Centreon, Cacti, HPNMN,
- Automation: Current automation relies on PowerShell, Usage of API through Python scripts, Terraform / Ansible

## 5.8 Guest OS

- Guest OS Mix: Windows Server (various versions), RHEL (various versions), other open-source Linux distributions. Estimated split 30% Windows, 30% Other Linux, 40% RHEL

## 5.9 Workload Profile

- A highly diverse mix of demanding enterprise workloads, including:
  - **Telco Network Functions (NFV):** [Mention specific types if known, e.g., EPC, IMS core components] often requiring specific performance tuning (NUMA, CPU pinning, SR-IOV - specify if confirmed).
  - **Large-Scale Databases:** Oracle, Microsoft SQL Server, MariaDB, PostgreSQL ...
  - Enterprise Applications: SAP, CRM Systems, Billing Platforms, ERP systems.
  - Virtual Desktop Infrastructure (VDI): CITRIX
  - General IT: Web Servers, Application Servers, File Servers, Domain Controllers, etc.
  - Top Critical Applications: Group Billing System, Pan-European SAP Instance, Regional Mobile Core VNFs, Global CRM Platform, ...

# 6 Technical Requirements for Proposed Solution

Vendors must complete the following Excel sheet, responding to each requirement using **"Fully Covered," "Partially Covered,"** or **"Not Covered."** Detailed explanations are **mandatory** for "Partially Covered" and strongly recommended for "Fully Covered" responses in the "Vendor Comments / Details" column. Reference specific service documentation or appendices where applicable.

*"Detailed\_requirements\_vvmare\_alternatives\_migration\_RFP"*

## 6.1 Core Hypervisor & Platform Architecture

1. **(TEC-HYP-001)** Describe the proposed platform architecture (e.g., kernel-based hypervisor, controller VMs, microservices management plane). Detail how this architecture ensures resilience and achieves high performance and scalability suitable for managing infrastructure distributed globally, for varying size of instances from small environments to very large DCs (dedicated or shared). Provide evidence or reference architectures demonstrating successful deployments at comparable scale.
2. **(TEC-HYP-002)** Confirm the platform supports installation on bare-metal commodity x86-64 server hardware. Provide a link to, or details of, the Hardware Compatibility List

(HCL). Describe the process for validating Client's existing hardware (primary vendors include Dell, HPE, Cisco – as described in Section 4) against the HCL.

3. **(TEC-HYP-003)** Specify the platform's documented scalability limits, including: Maximum physical RAM per host, Maximum logical CPUs per host, Maximum VMs per host, Maximum hosts per cluster (or equivalent management domain), and Maximum hosts/VMs managed by a single management instance/cluster.
4. **(TEC-HYP-004)** Specify the hypervisor's typical installation footprint (disk space required on host) and its typical runtime memory overhead per host.
5. **(TEC-HYP-005)** Describe the platform's support for CPU overcommitment (allocating more vCPUs than physical cores/threads). Explain the scheduling mechanism used.
6. **(TEC-HYP-006)** Describe the platform's support for Memory overcommitment (allocating more VM RAM than physical host RAM). Detail the mechanisms employed (e.g., memory ballooning, memory compression, hypervisor swapping, Transparent Page Sharing equivalent) and recommended best practices.

## 6.2 Compute Resource Management

1. **(TEC-CMP-001)** Confirm the platform supports live migration of running VMs between physical hosts without downtime or service interruption (equivalent to VMware vMotion). Describe any limitations (e.g., CPU family compatibility requirements, network constraints).
7. **(TEC-CMP-002)** Confirm the platform provides automated workload balancing based on host CPU and Memory utilization (equivalent to VMware DRS). Describe the algorithms used, the granularity of policy options (e.g., thresholds, aggression levels, automation levels - manual/partially/fully automated), and how resource priorities are handled.
8. **(TEC-CMP-003)** Describe the capability to create resource pools (or equivalent constructs) with configurable reservations, shares (priorities), and limits for CPU and Memory resources applicable to groups of VMs.
9. **(TEC-CMP-004)** Confirm support for defining VM-to-VM affinity ('keep together') and anti-affinity ('separate') rules, as well as VM-to-Host affinity/anti-affinity rules (required/preferred hosts).
10. **(TEC-CMP-005)** Describe the process for placing a host into maintenance mode. Confirm this process automatically live-migrates running VMs off the host to other available hosts in the cluster without downtime.
11. **(TEC-CMP-006)** Explain how the platform's resource scheduler handles diverse and demanding workloads co-existing within the same cluster (e.g., latency-sensitive Telco VNFs, high-IOPS Databases, CPU-intensive SAP, bursty VDI). Describe features ensuring fairness and adherence to resource reservations/limits/priorities.

## 6.3 Storage Capabilities

1. **(TEC-STO-001)** Confirm support for Fibre Channel (FC) and IP storage connectivity. Specify supported features (e.g., N\_Port ID Virtualization (NPIV), multi-pathing (MPIO) capabilities and policies).
12. **(TEC-STO-002)** Confirm support for iSCSI storage connectivity. Specify supported features (e.g., MPIO, Jumbo Frames support, CHAP authentication).
13. **(TEC-STO-003)** Confirm support for NFS storage connectivity. Specify supported versions (NFSv3 required, NFSv4.1+ preferred) and features (e.g., Kerberos authentication support, pNFS if applicable).
14. **(TEC-STO-004)** Confirm support for SMB3 storage connectivity (if applicable).
15. **(TEC-STO-005)** Describe the platform's construct for presenting and managing shared storage resources (equivalent to VMware datastores).
16. **(TEC-STO-006)** Confirm support for thin provisioning of virtual disks, allowing storage capacity to be allocated on demand.
17. **(TEC-STO-007)** Confirm support for thick provisioning of virtual disks (e.g., lazy-zeroed and/or eager-zeroed equivalents).
18. **(TEC-STO-008)** Describe the platform's mechanism for block-level space reclamation for thin-provisioned storage (equivalent to UNMAP/TRIM commands). Is this process automated or manually triggered?
19. **(TEC-STO-009)** Describe the native VM snapshot capabilities. Explain the underlying mechanism (e.g., redirect-on-write, copy-on-write), performance impact during snapshot existence, snapshot consolidation process, maximum number of snapshots per VM, and management interface.
20. **(TEC-STO-010)** Describe integration with storage array offload capabilities (equivalent to VMware VAAI) for operations like cloning, zeroing, and hardware-assisted locking. List supported storage vendor integrations or supported standards (e.g., T10 XCOPY).
21. **(TEC-STO-011)** Hyperconverged/Software-Defined Storage (HCI/SDS): State clearly whether the proposed platform includes a native, integrated Software-Defined Storage (SDS) / Hyperconverged Infrastructure (HCI) solution that pools local storage from cluster hosts. If Yes, respond to the following HCI-specific requirements. If No, state "Not Covered" for the following HCI requirements (TEC-STO-HCI-001 to 008).
22. **(TEC-STO-HCI-001)** If offering HCI/SDS: Describe the architecture (e.g., distributed storage services integrated into hypervisor kernel, requires dedicated Controller VMs). Detail minimum and recommended hardware requirements (disk types - SSD/NVMe/HDD ratios, network speed/latency requirements).
23. **(TEC-STO-HCI-002)** If offering HCI/SDS: Describe the available data resilience options (e.g., Replication Factor - RF2/RF3, Erasure Coding schemes like RAID-5/RAID-6 equivalents). Explain protection against disk and node failures.



24. **(TEC-STO-HCI-003)** If offering HCI/SDS: Detail available storage efficiency features (Deduplication, Compression). Specify if they operate inline or post-process, and describe the scope (e.g., cluster-wide, per-node, per-VM).
25. **(TEC-STO-HCI-004)** If offering HCI/SDS: Confirm the availability of Storage Quality of Service (QoS) capabilities. Describe how IOPS and/or bandwidth limits/reservations can be applied (e.g., per VM, per virtual disk).
26. **(TEC-STO-HCI-005)** If offering HCI/SDS: Describe how the solution scales capacity and performance. Can nodes and/or disks be added online without disruption?
27. **(TEC-STO-HCI-006)** If offering HCI/SDS: Describe the self-healing capabilities following a disk or node failure (e.g., automatic data rebuild process).
28. **(TEC-STO-HCI-007)** If offering HCI/SDS: Confirm support for creating storage policies or profiles that allow administrators to define VM storage characteristics (e.g., desired resilience level, performance tier, efficiency features).
29. **(TEC-STO-HCI-008)** If offering HCI/SDS: Provide performance benchmarks, reference architectures, or case studies demonstrating suitability for mixed demanding workloads (Telco, Databases, VDI, SAP) at scale.

## 6.4 Networking Capabilities (SDN Focus)

1. **(TEC-NET-001)** Describe the platform's native virtual switching capabilities (equivalent to vSphere Standard Switch / Distributed Switch).
2. **(TEC-NET-002)** Confirm the virtual switch supports standard features: 802.1Q VLAN tagging, multiple NIC teaming/bonding policies (e.g., active/standby, load-based teaming, LACP - IEEE 802.3ad).
3. **(TEC-NET-003)** Confirm the virtual switch supports traffic shaping capabilities (average bandwidth, peak bandwidth, burst size) for outbound traffic, and ideally inbound traffic.
4. **(TEC-NET-004)** Confirm platform-wide support for Jumbo Frames (MTU > 1500 bytes) on virtual and physical network interfaces used for VM traffic, storage traffic, and live migration.
5. **(TEC-NET-005)** Confirm platform-wide support for IPv6 addressing for host management interfaces and virtual machine network interfaces.
6. **(TEC-NET-SDN-001)** Software-Defined Networking (SDN): State clearly whether the proposed platform includes a comprehensive, mature, scalable, and native SDN solution providing functional parity with VMware NSX Data Center / NSX-T features used by the Client (Micro-segmentation, Routing, LB, VPN). If Yes, respond to the following SDN-specific requirements. If No, state "Not Covered" for the following SDN requirements (TEC-NET-SDN-002 to 010).
7. **(TEC-NET-SDN-002)** If offering SDN: Describe the overall architecture (e.g., overlay network technology used - VXLAN/Geneve, control plane architecture, data plane implementation, integration with physical underlay).

8. **(TEC-NET-SDN-003)** If offering SDN: Detail the distributed firewalling / micro-segmentation capabilities. Confirm support for L4 stateful firewalling between VMs. Describe L7 application identity firewalling capabilities if available. Detail the policy object model (e.g., based on VM identity, tags, IP addresses, security groups) and scalability for managing rules across thousands of VMs.
9. **(TEC-NET-SDN-004)** If offering SDN: Describe distributed logical routing capabilities (East-West traffic between virtual networks without hairpinning to physical routers). Describe centralized routing gateway capabilities (North-South traffic). Confirm support for dynamic routing protocols (BGP required, OSPF preferred) for peering with the physical network.
10. **(TEC-NET-SDN-005)** If offering SDN: Confirm support for Network Address Translation (NAT) services (SNAT, DNAT, potentially PAT).
11. **(TEC-NET-SDN-006)** If offering SDN: Describe native Load Balancing capabilities. Specify supported levels (L4 TCP/UDP, L7 HTTP/HTTPS), load balancing algorithms, health check methods, and SSL offload capabilities.
12. **(TEC-NET-SDN-007)** If offering SDN: Describe native VPN capabilities (IPsec site-to-site VPN required, SSL remote access VPN optional but desired).
13. **(TEC-NET-SDN-008)** If offering SDN: Detail how the SDN solution integrates with the physical network infrastructure (e.g., gateway connectivity, routing protocol peering, VLAN trunking).
14. **(TEC-NET-SDN-009)** If offering SDN: Describe support for network service insertion or service chaining, allowing integration with third-party virtual appliances (e.g., next-gen firewalls, intrusion detection systems).
15. **(TEC-NET-SDN-10)** If offering SDN: Detail the tools and capabilities provided for monitoring, troubleshooting, and analytics within the SDN fabric (e.g., flow monitoring/visualization - equivalent to vRealize Network Insight/Aria Operations for Networks, packet capture, port mirroring, latency monitoring).

## 6.5 Management, Operations & Automation

1. **(TEC-MGT-001)** Describe the centralized management interface(s). Confirm a comprehensive graphical user interface (GUI) is provided. Detail the availability, capabilities, and robustness of the command-line interface (CLI) and Application Programming Interface (API).
30. **(TEC-MGT-002)** Confirm the management platform architecture is proven to operate efficiently and reliably at the scale as described in Section 4. Describe the scaling architecture (e.g., use of federated management, regional instances, database scalability). Provide evidence or reference deployments.
31. **(TEC-MGT-003)** Confirm support for fine-grained Role-Based Access Control (RBAC) allowing definition of custom roles with specific permissions and IDP technologies such

as OAuth2 / OIDC. Describe integration capabilities with directory services (Microsoft Active Directory required, LDAP optional).

32. **(TEC-MGT-004)** Confirm the platform provides native, integrated capabilities equivalent to VMware Aria Operations (vROps) for comprehensive performance monitoring (real-time and historical metrics for compute, storage, network), health alerting, workload optimization recommendations, log management (vRLI), network management (vRNI) and capacity planning/forecasting. Describe key features and metrics available. Alternatively, if native parity is not claimed, describe built-in monitoring and the primary strategy for achieving advanced operations management (e.g., tight integration with specific third-party tools).
33. **(TEC-MGT-005)** Confirm the availability of customizable dashboards and reporting capabilities suitable for visualizing key metrics, inventory, capacity, and performance trends across a large, complex environment.
34. **(TEC-MGT-006)** Describe native multi-tenancy features equivalent to vCloud Director. Detail capabilities for Organization VDCs (or equivalent resource boundaries), user/group management per tenant, network isolation between tenants, service catalogs, and potential integration points for metering/showback/billing.
35. **(TEC-MGT-007)** Describe the platform's built-in automation capabilities for common tasks like VM provisioning, configuration updates, or scheduled tasks. Describe how these integrate with or complement external automation tools (see [Section 4.7](#)). Detail capabilities equivalent to Aria Automation for self-service portals, approval workflows, and infrastructure-as-code deployments.
36. **(TEC-MGT-008)** Confirm support for VM templating for rapid deployment. Describe the cloning process (e.g., linked clones, full clones). Detail support for guest OS customization during deployment (Sysprep for Windows, Cloud-init for Linux) and OS image factory capabilities like image catalogs, hardening/customization facilities etc.
37. **(TEC-MGT-009)** Describe the end-to-end process for patching and upgrading the platform components (hypervisors, management plane, SDN/HCI components). Confirm if the process is centralized, automated, supports orchestrated rolling upgrades to minimize downtime, and detail expected disruption levels during upgrades in large environments.
38. **(TEC-MGT-010)** The platform shall be delivered as containerized services. In particular, the hypervisor's control and management functions should run as containerized applications—integrated with Kubernetes to facilitate automated lifecycle management. This approach should enable seamless deployment, rolling updates, live patching, and scaling of both management functions and operational components, while guaranteeing zero or minimal service interruption.
39. **(TEC-MGT-011)** List of compatibility with cluster management platform and tools (like SUSE rancher for example)

## 6.6 High Availability & Resilience

1. **(TEC-HA-001)** Confirm the platform provides an automated VM restart mechanism in the event of a physical host failure (equivalent to vSphere HA). Describe the failure detection mechanism, typical failover time (detection + restart), configuration options (e.g., admission control policies, VM restart priorities), and handling of network/storage isolation scenarios.
40. **(TEC-HA-002)** Describe the high availability architecture for the management plane itself. How is resilience achieved against failure of management components or the hosts they run on?
41. **(TEC-HA-003)** Describe how the platform ensures resilience against network path failures impacting host management connectivity, storage connectivity (e.g., via MPIO), and VM network traffic (e.g., via NIC teaming).
42. **(TEC-HA-004)** If applicable (e.g., for HCI/SDS or management plane clustering), describe the cluster quorum mechanisms used to prevent split-brain scenarios during network partitions.
43. **(TEC-HA-005)** Describe any specific features or configuration options available to enhance resilience or performance determinism for demanding workloads like Telco VNFs (e.g., advanced NUMA awareness/controls, CPU pinning capabilities, network latency sensitivity tuning, rapid fault detection/failover mechanisms).

## 6.7 Backup, Recovery, and Disaster Recovery Integration

1. **(TEC-BKP-001)** Confirm the platform provides stable, documented APIs (functionally equivalent to VMware VADP) enabling third-party backup solutions to perform efficient, image-level backups of VMs (including Changed Block Tracking (CBT) equivalent functionality).
44. **(TEC-BKP-002)** List backup vendors whose solutions are certified or officially supported for integration with the proposed platform via the backup APIs. Prioritize confirmation for Veeam and Commvault. Provide links to relevant integration documentation or partner pages.
45. **(TEC-BKP-003)** Describe the platform's native capabilities for replicating VMs between sites or clusters (asynchronous replication required, synchronous optional). Specify achievable Recovery Point Objectives (RPOs) and any dependencies (e.g., licensing, network bandwidth/latency).
46. **(TEC-BKP-004)** Disaster Recovery Orchestration: State clearly whether the proposed platform includes a comprehensive, native DR orchestration solution providing functional parity with VMware SRM Enterprise. If Yes, respond to the following DR-specific requirements. If No, state "Not Covered" and describe the alternative strategy for achieving orchestrated DR (e.g., reliance on backup vendor replication/orchestration, third-party DR tools, scripting).

47. **(TEC-BKP-DR-001)** If offering DR orchestration: Confirm support for creating, managing, and automating failover and failback workflows (Recovery Plans).
48. **(TEC-BKP-DR-002)** If offering DR orchestration: Confirm support for executing non-disruptive DR tests without impacting production workloads.
49. **(TEC-BKP-DR-003)** If offering DR orchestration: Confirm support for customizing IP addresses and network settings for VMs during failover.
50. **(TEC-BKP-DR-004)** If offering DR orchestration: Confirm support for integrating custom scripts within recovery plans (e.g., pre/post power-on scripts).
51. **(TEC-BKP-DR-005)** If offering DR orchestration: Confirm availability of detailed reporting for both test and actual failover/failback events.
52. **(TEC-BKP-DR-006)** If offering DR orchestration: Describe how the solution scales to manage DR plans for potentially thousands of VMs across multiple sites or protection groups.
53. **(TEC-BKP-DR-007)** If offering DR orchestration: Describe integration capabilities with different replication sources (e.g., native platform replication, supported storage array-based replication).

## 6.8 Integration & Automation (API, Ecosystem)

1. **(TEC-INT-001)** Confirm the availability of a comprehensive, well-documented RESTful API covering all major platform management functions (VM lifecycle, host management, storage, networking, monitoring). Provide a link to the API documentation portal.
54. **(TEC-INT-002)** Confirm the availability of official Software Development Kits (SDKs) for common programming/scripting languages (e.g., Python, PowerShell, Go).
55. **(TEC-INT-003)** Confirm the availability of an official, supported Ansible collection (or equivalent modules) for automating platform configuration and management tasks.
56. **(TEC-INT-004)** Confirm the availability of an official, supported Terraform provider for managing platform resources as infrastructure-as-code.
57. **(TEC-INT-005)** Confirm the availability of a Container Storage Interface (CSI) driver allowing Kubernetes clusters (running inside or outside the platform) to consume persistent storage provided by the platform (especially relevant if HCI/SDS is offered).
58. **(TEC-INT-006)** Confirm the availability of a Cloud Provider Interface (CPI) allowing Kubernetes clusters running on the platform to integrate for node management, load balancer provisioning (if SDN LB available), etc.
59. **(TEC-INT-007)** Describe the ecosystem of validated or supported third-party integrations, particularly in areas critical to the Client (e.g., monitoring tools like Datadog/Splunk, security tools, backup solutions like Veeam/Commvault, automation platforms).

## 6.9 Guest OS Support

60. **(TEC-GOS-001)** Confirm full support, including performance-optimized drivers and management tools/agents, for current Microsoft Windows Server versions (2016, 2019, 2022).
61. **(TEC-GOS-002)** Confirm full support, including performance-optimized drivers (e.g., virtio) and management tools/agents, for current Red Hat Enterprise Linux (RHEL) versions (e.g., 7.x, 8.x, 9.x).
62. **(TEC-GOS-003)** Confirm full support (drivers, tools/agents) for other common Linux distributions used by the Client. List explicitly supported distributions and versions (e.g., Ubuntu 20.04 LTS, 22.04 LTS, CentOS Stream 8/9, SUSE SLES versions).
63. **(TEC-GOS-004)** Describe the process for installing and managing (including updates) guest tools/agents within supported virtual machines. Is there a centralized or automated update mechanism?

## 6.10 Performance & Scalability

1. **(TEC-PER-001)** Provide specific performance benchmarks, sizing guides, reference architectures, or verifiable case studies demonstrating the platform's suitability for Telco workloads (NFV) operating at scale. Include relevant metrics (e.g., packet throughput, latency).
64. **(TEC-PER-002)** Provide specific performance benchmarks, sizing guides, reference architectures, or verifiable case studies demonstrating the platform's suitability for large databases (Oracle/SQL Server). Include relevant metrics (e.g., IOPS, latency, throughput).
65. **(TEC-PER-003)** Provide specific performance benchmarks, sizing guides, reference architectures, or verifiable case studies demonstrating the platform's suitability for SAP HANA (confirm SAP HANA certification status if applicable).
66. **(TEC-PER-004)** Provide specific performance benchmarks, sizing guides, reference architectures, or verifiable case studies demonstrating the platform's suitability for large-scale VDI deployments. Specify tested user density, protocols (e.g., Blast, PCoIP, HDX), and workload types (e.g., task worker, knowledge worker).
67. **(TEC-PER-005)** Describe the scale-out architecture and identify any potential bottlenecks or practical limitations when scaling compute, storage (especially HCI/SDS), and networking (especially SDN control/data plane) components to the Client's target environment size.

# 7 Security & Compliance Requirements



1. **(TEC-SEC-001)** Describe the built-in security hardening features of the hypervisor and management platform. Provide references to security configuration guides or best practice documentation.
68. **(TEC-SEC-002)** Confirm platform support for Secure Boot for the hypervisor hosts (requires UEFI hardware).
69. **(TEC-SEC-003)** Confirm platform support for Secure Boot for virtual machines (UEFI Secure Boot).
70. **(TEC-SEC-004)** Confirm platform support for integrating with external Key Management Servers (KMS) using the KMIP standard (specify supported KMIP versions).
71. **(TEC-SEC-005)** Describe native VM encryption capabilities (at-rest). Specify the scope (e.g., VM configuration files, all virtual disks, specific virtual disks) and implementation details (e.g., requires KMS, encryption key management).
72. **(TEC-SEC-006)** If offering HCI/SDS: Describe native data-at-rest encryption capabilities for the storage layer. Specify scope (e.g., cluster-wide, per-disk) and implementation (e.g., requires KMS, requires Self-Encrypting Drives (SEDs)).
73. **(TEC-SEC-007)** Confirm whether management traffic between platform components (e.g., management server to hypervisor) is encrypted by default using strong cryptographic protocols (e.g., TLS 1.2+).
74. **(TEC-SEC-008)** Confirm whether VM live migration traffic is encrypted by default or available as a configurable option.
75. **(TEC-SEC-009)** Describe the platform's audit logging capabilities. Confirm that all significant administrative actions and system events are logged. Detail the log format, retention capabilities, and export options (Syslog export required).
76. **(TEC-SEC-010)** List any relevant platform security certifications obtained (e.g., FIPS 140-2 validation, Common Criteria EAL certification). Specify certifications relevant to Client's industry (e.g., Telco, Finance) if applicable.
77. **(TEC-SEC-011)** Describe available tools or integration points for security posture management, including vulnerability scanning of the platform components (e.g., API for scanners) and compliance checking against security benchmarks (e.g., built-in checks, integration with third-party compliance tools).
78. **(TEC-SEC-012)** Detail any specific security features or compliance alignments relevant to Telco industry standards or other regulations governing the Client's business.
79. **(TEC-SEC-013)** Describe your current Status on CRA requirements implementation as a manufacturer. Cybersecurity is considered in planning, design, development, production, delivery and maintenance phase (the main obligations introduced by the Act will apply from 11 December 2027).

## 8 Licensing & Subscription Model

1. **(TEC-LIC-001)** Provide an extremely clear, simple, and predictable description of the proposed licensing and/or subscription model. Specify the primary metric (e.g., per physical CPU socket, per physical core, per node, per VM, capacity-based) and the standard term length(s).
80. **(TEC-LIC-002)** Provide a detailed feature matrix explicitly showing which capabilities (e.g., Core Hypervisor, HA, Live Migration, Central Management, Basic Monitoring, SDN Features (list key ones), DR Orchestration, Advanced Monitoring/Analytics, HCI/SDS Features (list key ones), API Access, Specific Security Features) are included in the base license/subscription tier versus those requiring add-on licenses or higher-cost premium tiers. Transparency is critical.
81. **(TEC-LIC-003)** Confirm that the proposed licensing model enables a clear path to significant TCO reduction compared to the incumbent VMware stack, aligning with the Client's primary objective. (Detailed TCO comparison requested in Pricing Section).
82. **(TEC-LIC-004)** Clarify how ongoing support and maintenance costs are factored into the licensing/subscription model (e.g., included in subscription, percentage of license cost, separate line item per tier).
83. **(TEC-LIC-005)** Describe options available for ensuring long-term price predictability or protection (e.g., multi-year contract discounts, fixed rate options, capped increases).
84. **(TEC-LIC-006)** Confirm whether licenses/subscriptions are portable and can be reapplied if the underlying physical hardware is refreshed (assuming the new hardware is on the HCL).
85. **(TEC-LIC-007)** Is it possible to have fix price during the group contract duration? (avoid price increases during the contract period)

## 9 Migration Requirements & Methodology

### 9.1 Migration Tools & Technology

1. **(MIG-TOL-001)** Describe the proposed V2V (VMware-to-Vendor) migration tools (native or third-party). Confirm they are proven and scalable for the Client's environment size and complexity (multiple source versions). Provide evidence of scalability.
86. **(MIG-TOL-002)** Confirm tool support for Cold/Offline migration.
87. **(MIG-TOL-003)** Confirm tool support for Warm/Live migration with minimal downtime suitable for critical workloads. Describe the mechanism (e.g., background sync, CBT usage, final cutover process) and typical expected cutover duration/disruption per VM.
88. **(MIG-TOL-004)** Detail agentless vs. agent-based migration capabilities and when each method is used.
89. **(MIG-TOL-005)** Describe the automation level of the migration tools (e.g., bulk VM scheduling, network re-IP / maintaining existing IP addresses capabilities, automated guest tool installation post-migration).



90. **(MIG-TOL-006)** Confirm tool support for migrating VMs with complex configurations (e.g., multiple vDisks, specific network settings, potentially RDMS).
91. **(MIG-TOL-007)** Describe capabilities or strategies for migrating network configurations from NSX (firewall rules, segments, LB pools) to the proposed SDN solution. Detail level of automation vs. manual effort expected.
92. **(MIG-TOL-008)** Describe capabilities or strategies for migrating DR configurations from SRM (recovery plans, mappings) to the proposed DR orchestration solution. Detail level of automation vs. manual effort expected.

## 9.2 Migration Planning & Assessment

1. **(MIG-PLN-001)** Describe tools and/or professional services offered for comprehensive pre-migration assessment at scale (inventory, application dependency mapping, compatibility checks, right-sizing).
93. **(MIG-PLN-002)** Confirm the assessment process includes analysis of NSX and SRM configurations to inform the migration strategy for network/security policies and DR plans.
94. **(MIG-PLN-003)** Detail the proposed migration planning methodology, including defining migration waves/groups suitable for a large, global environment with diverse dependencies and risk profiles.
95. **(MIG-PLN-004)** Describe how technical prerequisites (network connectivity between sites, landing zone configuration in the target platform, bandwidth requirements) are identified, planned for, and validated.

## 9.3 Migration Execution Process

1. **(MIG-EXE-001)** Describe the step-by-step process for migrating a single VM and, critically, managing automated bulk VM migrations reliably and efficiently at scale.
96. **(MIG-EXE-002)** Describe how migration progress is monitored and reported across large migration waves. What metrics are tracked?
97. **(MIG-EXE-003)** Describe the process for handling migration failures (individual VM, batch), performing retries, and troubleshooting migration issues.
98. **(MIG-EXE-004)** Provide estimated migration speed or throughput per VM or per batch based on typical network conditions.

## 9.4 Validation & Cutover Support

1. **(MIG-VAL-001)** Describe the tools or standard procedures used for post-migration validation (e.g., automated VM power-on checks, network connectivity tests, assisting Client with application-level validation triggers).
99. **(MIG-VAL-002)** Detail the proposed cutover strategy options (e.g., pilot groups, phased cutover by application/site/affiliate, rollback triggers).

100. **(MIG-VAL-003)** Describe the standard rollback plan and process in case of critical issues discovered during or immediately after cutover for a migration wave.
101. **(MIG-VAL-004)** Detail the level and availability (e.g., dedicated team, 24x7 availability) of vendor technical support specifically provided during critical migration execution and cutover windows ("hypercare").

## 9.5 Professional Services for Migration

1. **(MIG-PSS-001)** Provide a detailed description of the vendor's professional services portfolio specifically designed for large-scale, complex VMware replacement and migration projects.
102. **(MIG-PSS-002)** Propose a clear Roles and Responsibilities (RACI) matrix covering the entire migration lifecycle (Assessment, Planning, Design, Tooling Setup, Execution, Validation, Cutover, Hypercare), delineating Vendor vs. Client tasks.
103. **(MIG-PSS-003)** Provide a sample high-level migration project plan indicating typical phases, durations, dependencies, and milestones applicable to an environment of this scale.
104. **(MIG-PSS-004)** Describe the typical experience level, qualifications, and relevant certifications of the migration architects, consultants, and project managers proposed for this engagement.
105. **(MIG-PSS-005)** Describe knowledge handover process ensuring that clients will be able to reuse the approach independently after the migration plan concludes.
106. **(MIG-PSS-005)** How does the knowledge handover process look to ensure that clients will be able to reuse the approach independently after the migration plan concludes?

## 10 Support and Maintenance Services

1. **(SUP-TIE-001)** Describe available support tiers (e.g., Standard, Premium, Enterprise). Clearly detail the service components, features, and SLA differences included in each tier relevant to the Client's needs.
107. **(SUP-SLA-001)** Detail the Service Level Agreements (SLAs) for initial support response times based on incident severity (provide definitions for Critical, High, Medium, Low severity or refer to the ones described in Section 11.1).
108. **(SUP-LNG-001)** Confirm guaranteed support availability in English and French. Describe how multilingual support is provided.
109. **(SUP-GEO-001)** Confirm ability to provide effective support across Client's key operating regions (EU and MEA time zones). Describe the operational model (e.g., follow-the-sun, regional centers).

110. **(SUP-ACC-001)** Describe the methods available for accessing technical support (e.g., Web portal, Phone, Email). Confirm requirements for portal access and mandatory phone access for critical issues.
111. **(SUP-ESC-001)** Provide details on the documented escalation procedure (technical and management) for issues not being resolved satisfactorily or for SLA breaches. Include contact methods/points.
112. **(SUP-TAM-001)** Describe the availability and role of a dedicated Technical Account Manager (TAM) or equivalent senior designated support contact. Specify if included in proposed tiers or optional.
113. **(SUP-UPD-001)** Describe the process for delivering platform software updates, patches, and security fixes. Detail notification methods, release cadences, and recommended deployment practices.
114. **(SUP-UPD-002)** Describe how vendor commercial maintenance and support can remain fully effective and applicable, even when the hypervisor is managed through Orange automation and lifecycle tooling, without any limitations or voiding of contractual support obligations.
115. **(SUP-UPD-003)** Describe how many versions are supported instead of the softwares actual versions (N-1, N-2,N-3 ?). What is the average support period of a version before being not supported ?. What is the average time between two main version upgrades ? Is it possible to upgrade from one version to another version without upgrading to intermediate version ?

## 11 Vendor Qualifications and Experience Requirements

1. **(VEN-OVR-001)** Provide Company Overview: Brief history, years in business, financial stability overview, number of employees, primary focus (virtualization, cloud, HCI, etc.).
116. **(VEN-EXP-001)** Demonstrate significant, verifiable experience replacing large-scale (thousands of hosts / 100k+ cores), complex VMware environments.
117. **(VEN-EXP-002)** Detail experience specifically with environments involving NSX and/or SRM replacement as part of the VMware migration.
118. **(VEN-EXP-003)** Detail experience supporting demanding workloads (Telco, Finance/Billing, SAP, Large DBs, VDI) on your platform post-migration from VMware.
119. **(VEN-EXP-004)** Provide anonymized case studies of comparable VMware replacement projects, highlighting challenges, solutions, migration approach, and outcomes (especially TCO savings).
120. **(VEN-MIG-001)** Detail vendor's specific experience performing V2V migrations from multiple legacy vSphere versions (v4-v8) to your current platform version.

121. **(VEN-STF-001)** Describe the qualifications, certifications (on vendor's own platform), and average years of experience for technical staff involved in: platform implementation, migration, support.
122. **(VEN-REF-001)** Provide at least three (3) large enterprise client references (preferably global companies, Telco/Finance if possible) who have completed a similar large-scale VMware replacement project using your platform and migration services.

## 12 Training Requirements

2. **(TRN-OFF-001)** Describe the portfolio of available training offerings covering the proposed platform. Ensure topics include: Platform Administration, Day-to-Day Operations, Troubleshooting, Networking/SDN Management, Storage/HCI Management, and DR Orchestration Management.
123. **(TRN-OFF-002)** Detail the available training formats (e.g., instructor-led virtual, instructor-led classroom, self-paced online courses, hands-on labs, official documentation).
124. **(TRN-OFF-003)** Describe any official certification programs available for the proposed platform, including levels and specialization tracks relevant to administration and operations.
125. **(TRN-PLN-001)** Propose a recommended training plan, including specific courses, formats, duration, and associated costs, for the Client's technical team.

## 13 Details of Service Level Agreements (SLAs)

This section outlines the minimum Service Level Agreements (SLAs) that the selected Vendor must adhere to when providing support services for the Client's environment as defined in this RFP. All timeframes are measured from the moment an incident is successfully logged by the Client via an agreed-upon method and acknowledged by the Vendor's system or support personnel.

### 13.1 Incident Severity Level Definitions

The Client and Vendor will mutually agree on the severity level assigned to an incident based on business impact. The following definitions will serve as the standard classification guide:

- **Severity 1 (Critical / Urgent):**
  - **Definition:** Critical production system outage or major business disruption affecting a significant number of users or critical business processes. Core virtualisation infrastructure components are down or severely degraded, preventing essential operations. No immediate solution or workaround is available.

- o **Example:** Entire production cluster offline, unable to power on critical VMs; management capabilities inaccessible preventing management of the environment; critical security breach related to a virtualisation component failure.
- **Severity 2 (High):**
  - o **Definition:** Significant service degradation or potential disruption impacting important business functions or a group of users. A core virtualisation feature is malfunctioning, or a redundant component has failed, increasing risk. A workaround may exist, but it is difficult or impractical.
  - o **Example:** failures preventing host maintenance; single host failure in a cluster where HA fails to restart VMs automatically; performance degradation significantly impacting a key application; failure of a redundant network path managed by the solution.
- **Severity 3 (Medium):**
  - o **Definition:** Minor service degradation affecting a limited number of users or non-critical business processes. A non-critical feature is malfunctioning, or a cosmetic issue exists. A reasonable workaround is available. Includes requests for configuration guidance or non-urgent troubleshooting.
  - o **Example:** Issues with snapshot consolidation on a non-critical VM; questions about configuring a specific vSwitch setting; performance issue on a development environment VM; log analysis request for a past minor event.
- **Severity 4 (Low):**
  - **Definition:** General inquiries, requests for information, documentation clarification, or planned configuration assistance where no service impact exists.
  - **Example:** Request for documentation on a specific feature; general question about best practices; planned assistance needed for a future configuration change.

## 13.2 Support Availability & Coverage

- **Severity 1 Availability:** Technical support for Severity 1 incidents must be available 24 hours per day, 7 days per week, 365 days per year (24x7x365).
- **Severity 2-4 Availability:** Technical support for Severity 2, 3, and 4 incidents must be available, at a minimum, during Standard Business Hours, defined as covering the Client's primary operating time zones across Europe (EU) and Middle East & Africa (MEA). Vendor should describe how they cover these time zones.
- **Language Support:** Support personnel interacting with the Client for all severity levels must be fluent in English. Support personnel capable of interacting fluently in French must also be available.
- **Support Channels:** Support must be accessible via Vendor's Web Portal, email, Phone, videoconference tool. Phone access is mandatory for Severity 1 incidents.

### 13.3 Response Time SLAs

"Response Time" is defined as the time elapsed between the Client successfully logging the incident via an agreed channel and the Vendor acknowledging the request AND assigning a qualified engineer who begins active troubleshooting or information gathering.

- Severity 1 (Critical): Maximum Initial Response Time: 30 minutes
- Severity 2 (High): Maximum Initial Response Time: 2 hours 30 minutes
- Severity 3 (Medium): Maximum Initial Response Time: 4 hours 30 minutes
- Severity 4 (Low): Maximum Initial Response Time: 8 hours 30 minutes

### 13.4 Communication Plan

During active troubleshooting for open incidents, the Vendor must provide regular status updates to the Client via the ticketing system or other agreed-upon channels. Minimum update frequencies are:

- **Severity 1 (Critical):** Updates every **1 hour** or upon significant findings/changes.
- **Severity 2 (High):** Updates every **4 Standard Business Hours** or upon significant findings/changes.
- **Severity 3 (Medium):** Updates every **1 Business Day** or upon significant findings/changes.
- **Severity 4 (Low):** Updates provided as information becomes available or resolution is achieved.

### 13.5 SLA Reporting

The vendor must provide monthly reports detailing performance against the Response Time SLAs for all incidents handled during the reporting period. Reports should show incidents counts by severity, average response times, and percentage of incidents meeting SLA targets.

## 14 Service Management & Reporting Procedures

This section details the mandatory procedures and minimum system requirements for managing and reporting on technical support incidents related to the virtualisation environment covered under the agreement. These procedures are designed to ensure timely, efficient, and transparent service delivery aligned with the Service Level Agreements (SLAs) defined in Section 12. The Vendor's adherence to these procedures is essential.

## 14.1 Incident Logging & Submission Channels

The Client will initiate support requests (Incidents) using the following Vendor-provided channels:

- **Vendor Web Portal (Preferred):** A secure online portal must be available for logging all incidents (Severity 1-4), tracking status, viewing communication history, and accessing relevant documentation.
- **Dedicated Phone Number (Mandatory for Severity 1):** A dedicated, toll-free (or regionally appropriate) phone number must be available 24x7x365 specifically for logging **Severity 1 (Critical)** incidents to ensure immediate attention. Phone support should also be available during Standard Business Hours for other severities if needed.
- **Email (Optional/Secondary):** Email may be used for logging low-priority (Severity 4) requests or for follow-up communication on existing tickets, but SLA response times will be measured from ticket creation in the Vendor's system.

## 14.2 Ticketing System Requirements

The Vendor must utilize a professional IT Service Management (ITSM) or ticketing system to log, track, manage, and report on all Client incidents. The system must:

- Assign a unique reference number (Ticket ID) to each incident upon creation.
- Record timestamps for incident creation, status changes, communications, and resolution.
- Allow for categorization by Severity Level (as defined in Section 12.1), status (e.g., New, Assigned, In Progress, Pending Client, Resolved, Closed), and potentially component type.
- Provide Client users with secure web portal access to:
  - View all their affiliate's open and recently closed incidents.
  - Check the current status and assigned engineer (if applicable).
  - Review the full communication history for each ticket.
  - Add updates or attachments to existing tickets.
- Support email notifications to the Client contact(s) upon key status changes (e.g., ticket creation, resolution).

### 14.3 Incident Classification & Prioritization

- Upon receipt of a new incident, the Vendor support team will validate the details and classify the incident according to the **Severity Level Definitions provided in Section 12.1** .
- The Vendor will prioritize work based on the assigned Severity Level, adhering to the **Response Time SLAs defined in Section 12.3**.
- If the Client disagrees with the Vendor's initial severity classification, they should immediately raise this concern with the assigned engineer or via the defined escalation path (Section 13.5) for review and potential adjustment based on documented business impact.

### 14.4 Incident Handling & Communication Workflow

- The Vendor must acknowledge receipt of the incident and assign a qualified engineer to begin work within the **Response Time SLAs** defined in Section 12.3.
- All significant troubleshooting steps, findings, communications (internal and external), and proposed solutions/workarounds must be documented within the incident ticket record in the ticketing system.
- The assigned Vendor engineer will provide regular status updates to the Client contact(s) via the ticketing system or other agreed channels, adhering to the minimum frequencies defined in the Communication Plan (Section 12.4) based on incident severity.
- Communication must be clear, concise, timely, and professional.
- Resolution or workaround implementation steps must be clearly documented in the ticket before marking an incident as Resolved. The ticket may be closed after Client confirmation or a pre-defined period of inactivity following resolution.

### 14.5 Escalation Procedures

- The Vendor must provide clearly documented procedures for both technical and management escalation:
- **Technical Escalation:** Define the internal process for escalating complex technical issues from initial support tiers to more senior engineers, specialists (e.g., certified experts), or engineering teams within the Vendor organization. This internal process should be transparent to the Client upon request.
- **Management Escalation:** Provide a clear, documented path for the Client to escalate concerns regarding:
  - o Lack of progress on an incident (especially high severity).



- o Dissatisfaction with the quality of support or communication.
- o Disagreement on incident severity classification.
- o Failure to meet agreed-upon SLAs.
- o This path must include specific contact points (e.g., Duty Manager, Support Manager, Account Manager) with contact details (phone number, email alias) accessible during relevant support hours (including 24x7 for critical issues).

## 15 Proposal Evaluation Criteria

Proposals will be evaluated based on a combination of factors, including, but not limited to:

- **TCO Reduction & Pricing Model (Very High Priority):** Clarity, predictability, and quantified cost savings compared to VMware. Completeness of pricing.
- **Migration Plan & Capability (Very High Priority):** Demonstrated ability, tools, and experience to handle the scale, complexity (NSX, SRM, multi-version vSphere), and critical workloads of the Client's environment with minimal risk/disruption.
- **Technical Fit & Feature Parity (High Priority):** Meeting requirements, especially for SDN (NSX parity), DR Orchestration (SRM parity), Scalability, Performance (for specific workloads), and Management at Scale.
- **Vendor Viability & Experience (High Priority):** Proven track record in large-scale VMware replacements, financial stability, relevant industry experience (Telco), and strong client references.
- **Support & Maintenance:** Alignment with Client's operational needs (SLAs, language, geography).
- **Ease of Use & Management:** Simplicity and efficiency for operations on a scale and level of automation
- **Contract Terms:** Reasonableness and risk allocation.

## 16 Appendices

Vendors may include relevant appendices, such as:

- Detailed technical specifications. Sample migration plan. Reference architecture diagrams. Standard Contract Documents (Required). Security/Compliance certifications. Case studies. TCO calculation worksheet. HCL details.